## Computer Science & Information Technology (CS) :

1. **Engineering Mathematics :**

   - **Discrete Mathematics**:
     - Propositional and first-order logic
     - Sets, relations, functions, partial orders, and lattices
     - Monoids and groups
     - Graphs: connectivity, matching, coloring
     - Combinatorics: counting, recurrence relations, generating functions

   - **Linear Algebra**:
     - Matrices, determinants
     - System of linear equations
     - Eigenvalues and eigenvectors
     - LU decomposition

   - **Calculus**:
     - Limits, continuity, and differentiability
     - Maxima and minima
     - Mean value theorem
     - Integration

   - **Probability and Statistics**:
     - Random variables
     - Distributions (uniform, normal, exponential, Poisson, binomial)
     - Mean, median, mode, and standard deviation
     - Conditional probability and Bayes' theorem

2. **Digital Logic**
   - Boolean algebra
   - Combinational and sequential circuits
   - Minimization
   - Number representations and computer arithmetic (fixed and floating point)

3. **Computer Organization and Architecture**
   - Machine instructions and addressing modes
   - ALU, data path, and control unit
   - Instruction pipelining and pipeline hazards
   - Memory hierarchy: cache, main memory, secondary storage
   - I/O interface (interrupt and DMA mode)

4. **Programming and Data Structures**
    - Programming in C
    - Recursion
    - Arrays, stacks, queues, linked lists, trees
    - Binary search trees, binary heaps, graphs

5. **Algorithms**
    - Searching, sorting, hashing
    - Asymptotic worst-case time and space complexity
    - Algorithm design techniques: greedy, dynamic programming, divide-and-conquer
    - Graph traversals, minimum spanning trees, shortest paths

6. **Theory of Computation**
    - Regular expressions and finite automata
    - Context-free grammars and push-down automata
    - Regular and context-free languages, pumping lemma
    - Turing machines and undecidability

7. **Compiler Design**
    - Lexical analysis, parsing, syntax-directed translation
    - Runtime environments
    - Intermediate code generation
    - Local optimization, data flow analyses (constant propagation, liveness analysis, common subexpression elimination)

8. **Operating System**
    - System calls, processes, threads, inter-process communication, concurrency, and synchronization
    - Deadlock
    - CPU and I/O scheduling
    - Memory management and virtual memory
    - File systems

9. **Databases**
    - ER-model
    - Relational model: relational algebra, tuple calculus, SQL
    - Integrity constraints, normal forms
    - File organization, indexing (e.g., B and B+ trees)
    - Transactions and concurrency control

10. **Computer Networks**
    - Concept of layering: OSI and TCP/IP Protocol Stacks
    - Basics of packet, circuit, and virtual circuit switching
    - Data link layer: framing, error detection, Medium Access Control, Ethernet bridging
    - Routing protocols: shortest path, flooding, distance vector, and link state routing
    - Fragmentation and IP addressing (IPv4, CIDR notation)
    - Basics of IP support protocols (ARP, DHCP, ICMP)
    - Network Address Translation (NAT)
    - Transport layer: flow control, congestion control, UDP, TCP, sockets
    - Application layer protocols: DNS, SMTP, HTTP, FTP, Email

## General Aptitude (GA) :

1. **Verbal Aptitude**:
    - Basic English grammar: tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech.
    - Basic vocabulary: words, idioms, and phrases in context.
    - Reading and comprehension.
    - Narrative sequencing.

2. **Quantitative Aptitude**:
    - Data interpretation: data graphs (bar graphs, pie charts, and other graphs representing data), 2- and 3-dimensional plots, maps, and tables.
    - Numerical computation and estimation: ratios, percentages, powers, exponents, logarithms, permutations and combinations, and series.
    - Mensuration and geometry.
    - Elementary statistics and probability.

3. **Analytical Aptitude**:
    - Logic: deduction and induction.
    - Analogy.
    - Numerical relations and reasoning.

4. **Spatial Aptitude**:
    - Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping.
    - Paper folding, cutting, and patterns in 2 and 3 dimensions.